observation delay study — highway; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]
bimodal / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bimodal / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bimodal / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bimodal / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
exponential / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
exponential / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
exponential / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
exponential / 0	MPC / actor-free	0.0 [0.0, 16.1]	10.0 [2.8, 30.1]
triangular / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
triangular / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
triangular / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
triangular / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]

n=20 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.

observation delay study — merge; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]
bimodal / 0	LANS / 42	10.0 [2.8, 30.1]	0.0 [0.0, 16.1]
bimodal / 0	LANS / 43	5.0 [0.9, 23.6]	0.0 [0.0, 16.1]
bimodal / 0	LANS / 44	10.0 [2.8, 30.1]	0.0 [0.0, 16.1]
bimodal / 0	MPC / actor-free	0.0 [0.0, 16.1]	5.0 [0.9, 23.6]
bursty / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bursty / 0	MPC / actor-free	0.0 [0.0, 16.1]	5.0 [0.9, 23.6]
exponential / 0	LANS / 42	10.0 [2.8, 30.1]	0.0 [0.0, 16.1]
exponential / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
exponential / 0	LANS / 44	10.0 [2.8, 30.1]	0.0 [0.0, 16.1]
exponential / 0	MPC / actor-free	0.0 [0.0, 16.1]	65.0 [43.3, 81.9]
triangular / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
triangular / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
triangular / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
triangular / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
uniform / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]

n=20 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.

observation delay study — intersection; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]
bimodal / 0	LANS / 42	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bimodal / 0	LANS / 43	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bimodal / 0	LANS / 44	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]
bimodal / 0	MPC / actor-free	0.0 [0.0, 16.1]	60.0 [38.7, 78.1]
bursty / 0	LANS / 42	5.0 [0.9, 23.6]	0.0 [0.0, 16.1]
bursty / 0	LANS / 43	5.0 [0.9, 23.6]	0.0 [0.0, 16.1]
bursty / 0	LANS / 44	0.0 [0.0, 16.1]	5.0 [0.9, 23.6]
bursty / 0	MPC / actor-free	0.0 [0.0, 16.1]	60.0 [38.7, 78.1]
exponential / 0	LANS / 42	0.0 [0.0, 16.1]	40.0 [21.9, 61.3]
exponential / 0	LANS / 43	0.0 [0.0, 16.1]	30.0 [14.5, 51.9]
exponential / 0	LANS / 44	0.0 [0.0, 16.1]	55.0 [34.2, 74.2]
exponential / 0	MPC / actor-free	0.0 [0.0, 16.1]	100.0 [83.9, 100.0]
triangular / 0	LANS / 42	5.0 [0.9, 23.6]	0.0 [0.0, 16.1]
triangular / 0	LANS / 43	5.0 [0.9, 23.6]	0.0 [0.0, 16.1]
triangular / 0	LANS / 44	5.0 [0.9, 23.6]	0.0 [0.0, 16.1]
triangular / 0	MPC / actor-free	0.0 [0.0, 16.1]	25.0 [11.2, 46.9]
uniform / 0	LANS / 42	30.0 [14.5, 51.9]	0.0 [0.0, 16.1]
uniform / 0	LANS / 43	20.0 [8.1, 41.6]	0.0 [0.0, 16.1]
uniform / 0	LANS / 44	20.0 [8.1, 41.6]	0.0 [0.0, 16.1]
uniform / 0	MPC / actor-free	0.0 [0.0, 16.1]	0.0 [0.0, 16.1]

n=20 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.

observation delay study — roundabout; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]
bimodal / 0	LANS / 42	40.0 [21.9, 61.3]	40.0 [21.9, 61.3]
bimodal / 0	LANS / 43	45.0 [25.8, 65.8]	45.0 [25.8, 65.8]
bimodal / 0	LANS / 44	35.0 [18.1, 56.7]	50.0 [29.9, 70.1]
bimodal / 0	MPC / actor-free	0.0 [0.0, 16.1]	90.0 [69.9, 97.2]
bursty / 0	LANS / 42	45.0 [25.8, 65.8]	10.0 [2.8, 30.1]
bursty / 0	LANS / 43	55.0 [34.2, 74.2]	15.0 [5.2, 36.0]
bursty / 0	LANS / 44	85.0 [64.0, 94.8]	5.0 [0.9, 23.6]
bursty / 0	MPC / actor-free	20.0 [8.1, 41.6]	45.0 [25.8, 65.8]
exponential / 0	LANS / 42	0.0 [0.0, 16.1]	80.0 [58.4, 91.9]
exponential / 0	LANS / 43	10.0 [2.8, 30.1]	65.0 [43.3, 81.9]
exponential / 0	LANS / 44	0.0 [0.0, 16.1]	85.0 [64.0, 94.8]
exponential / 0	MPC / actor-free	0.0 [0.0, 16.1]	100.0 [83.9, 100.0]
triangular / 0	LANS / 42	80.0 [58.4, 91.9]	5.0 [0.9, 23.6]
triangular / 0	LANS / 43	70.0 [48.1, 85.5]	10.0 [2.8, 30.1]
triangular / 0	LANS / 44	95.0 [76.4, 99.1]	5.0 [0.9, 23.6]
triangular / 0	MPC / actor-free	35.0 [18.1, 56.7]	30.0 [14.5, 51.9]
uniform / 0	LANS / 42	100.0 [83.9, 100.0]	0.0 [0.0, 16.1]
uniform / 0	LANS / 43	90.0 [69.9, 97.2]	0.0 [0.0, 16.1]
uniform / 0	LANS / 44	85.0 [64.0, 94.8]	0.0 [0.0, 16.1]
uniform / 0	MPC / actor-free	95.0 [76.4, 99.1]	0.0 [0.0, 16.1]

n=20 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.